

Exploratory Data Analysis of IPL Matches (2008–2024)

Project Report

Dataset: IPL Match-Level Records, 2008–2024
Exploratory Data Analysis using Python

Abstract

This project presents an exploratory analysis of 1,095 IPL match records represented by 20 variables. The analysis examines season-wise match distribution, team participation and wins, toss decisions, venues, match outcomes, winning margins, and special match situations. Python with Pandas, NumPy, Matplotlib and Jupyter Notebook was used for data handling and visualization. The analysis identifies descriptive patterns in the dataset without attempting to predict future match results.

1. Introduction

The Indian Premier League (IPL) is a professional Twenty20 cricket competition that produces extensive match-level data. This project applies Exploratory Data Analysis (EDA) to an IPL dataset covering 2008–2024 to summarize season distribution, team participation, toss decisions, venues, match outcomes and winning margins.

2. Problem Statement

The dataset contains many records and variables, making broad patterns difficult to identify from individual matches. EDA provides a systematic approach to inspect data quality, summarize variables and visualize distributions.

3. Objectives

- Understand dataset structure and quality.
- Examine season-wise match distribution.
- Analyze team participation and wins.
- Study toss decisions and match outcomes.
- Identify frequently represented venues.
- Analyze result types and winning margins.
- Examine special match situations.
- Communicate findings through visualizations.

4. Dataset Description

The dataset contains 1,095 records and 20 variables, with 17 season labels, 58 venues and 36 cities after documented cleaning.

5. Dataset Summary

Metric	Value
Matches	1,095
Variables	20
Season labels	17
Venues	58
Cities	36
Duplicate rows	0
Toss: field	704
Toss: bat	391
Wins by runs	498
Wins by wickets	578
Ties	14
No results	5

Metric	Value
Super overs	14
D/L matches	21

6. Data Preprocessing and Cleaning

Dates were converted to datetime where possible. Historical franchise names were standardized. Missing city values were filled only where the venue directly identified Dubai or Sharjah. No-result winners and genuine missing result margins were preserved rather than artificially imputed.

7. Exploratory Data Analysis and Visualizations

7.1 Season-wise Match Distribution

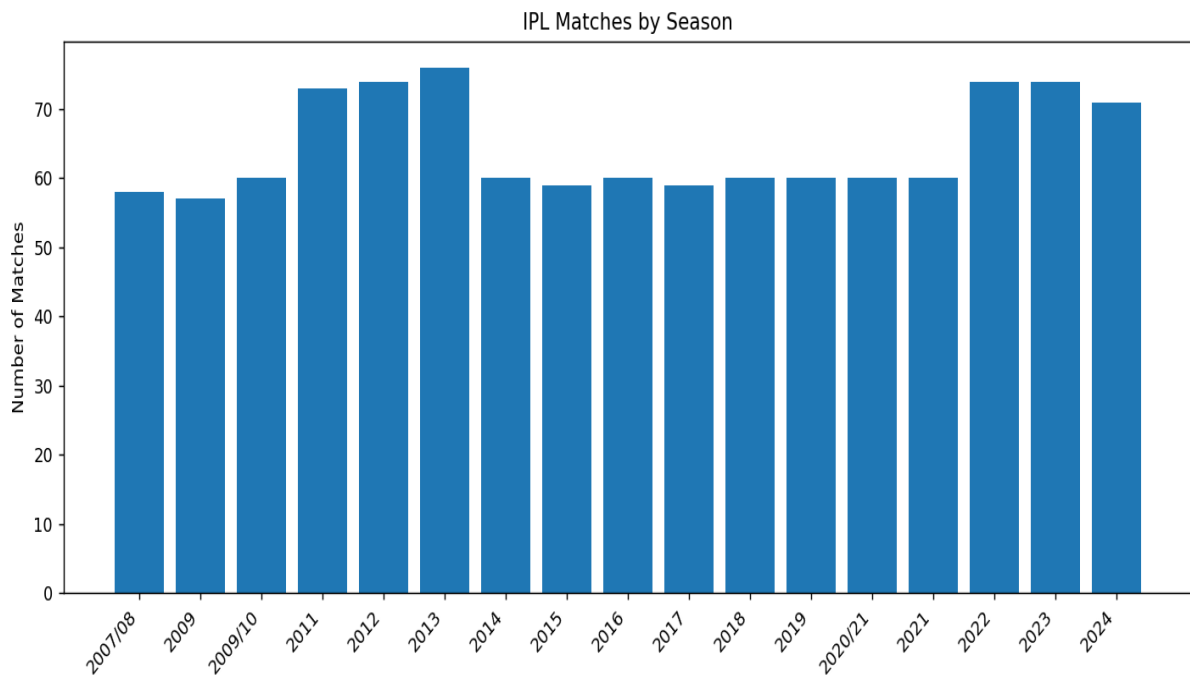


Figure 7.1: Season-wise Match Distribution

This chart shows how the number of recorded matches varies across season labels. It provides a high-level view of the tournament's match volume over the study period.

7.2 Team Winning Performance

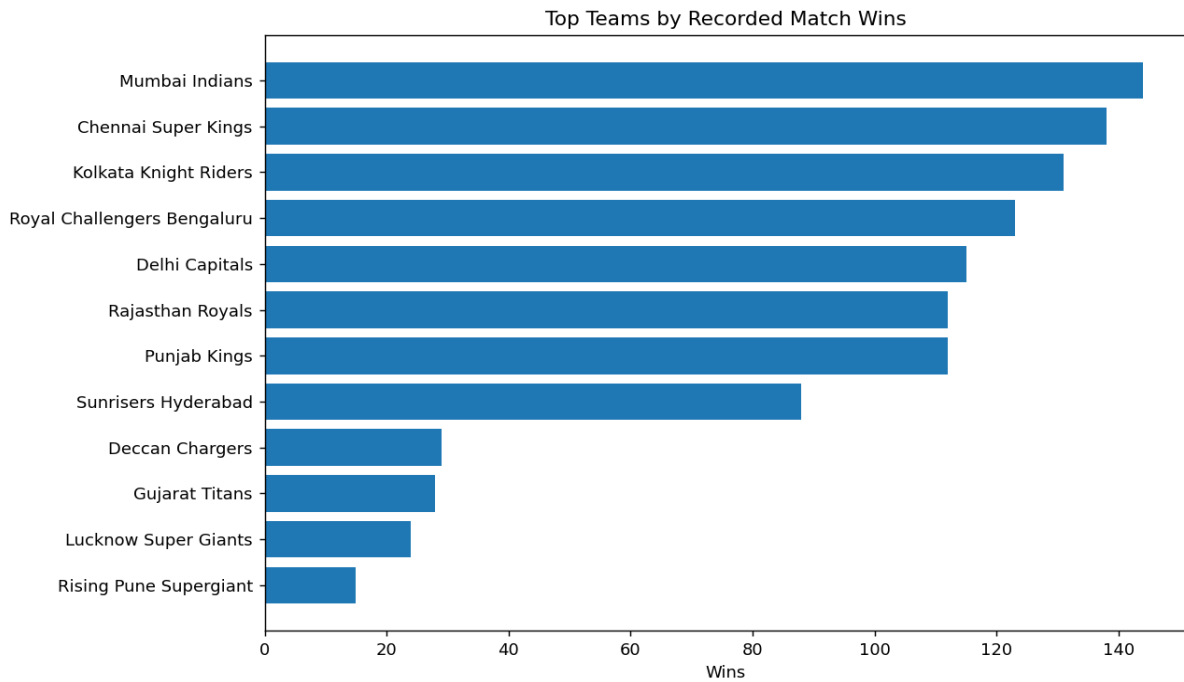


Figure 7.2: Team Winning Performance

The chart compares recorded match wins among teams. Mumbai Indians appears at the top of the ranking within this dataset, with 144 recorded wins.

7.3 Team Match Appearances

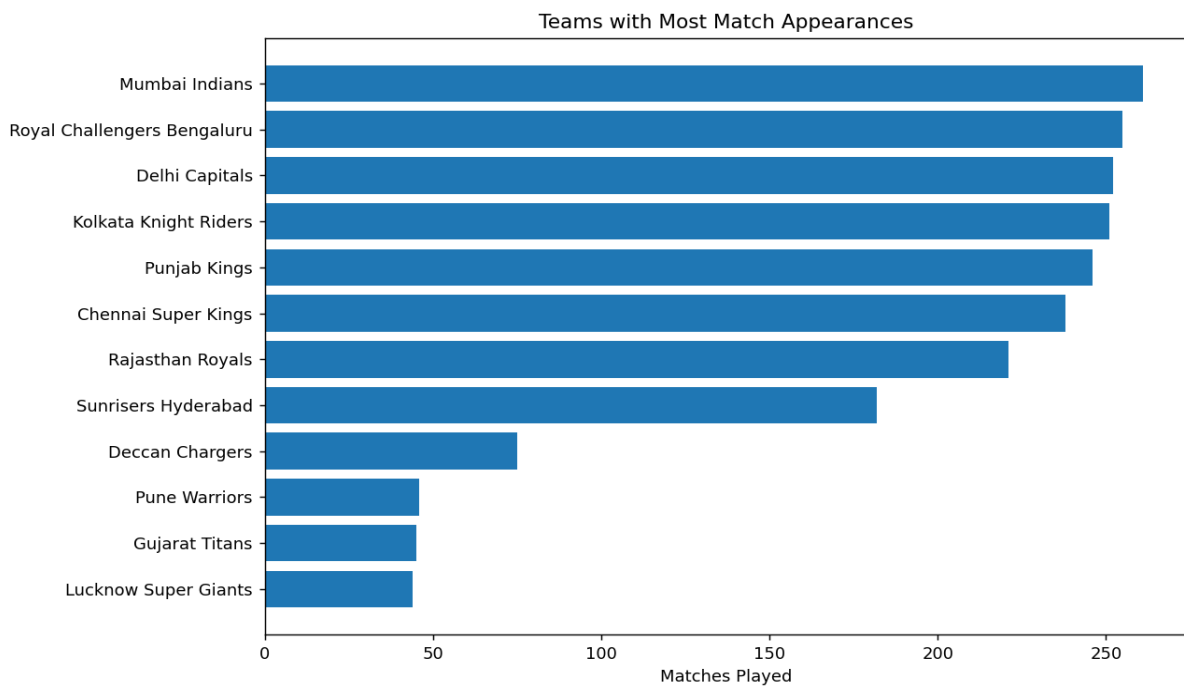


Figure 7.3: Team Match Appearances

This visualization compares team participation. Mumbai Indians has the largest number of recorded appearances (261).

7.4 Toss Decision Distribution

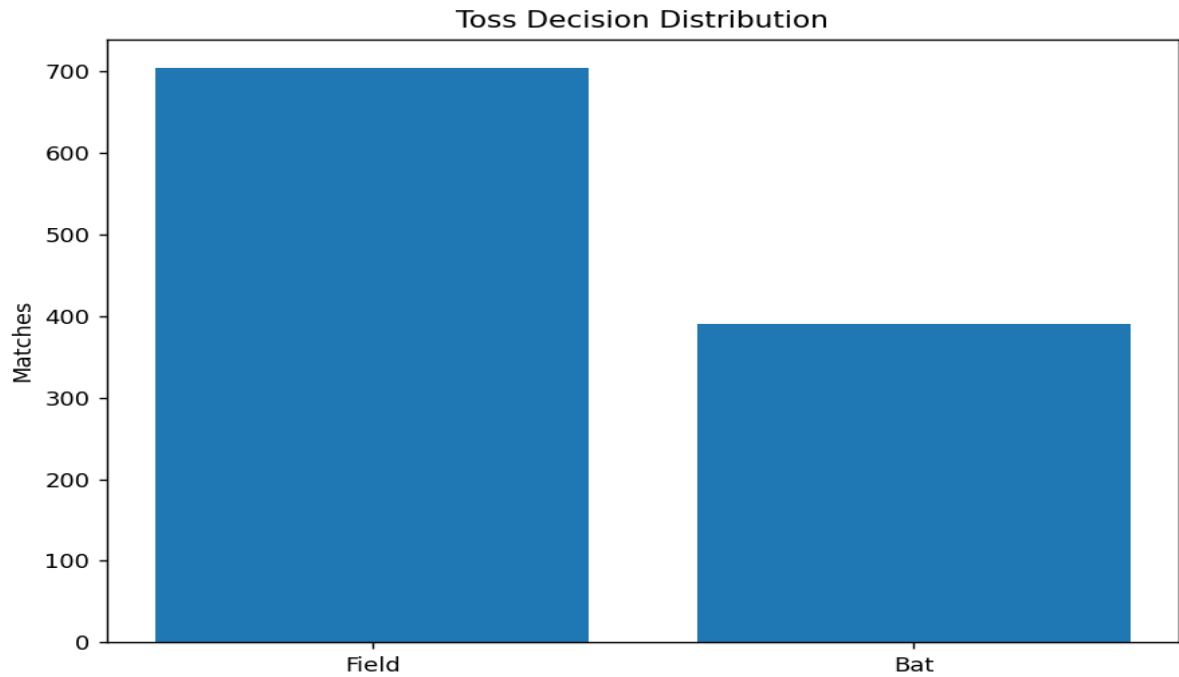


Figure 7.4: Toss Decision Distribution

Fielding was selected in 704 matches compared with 391 matches in which batting was selected. The chart summarizes the historical distribution of toss decisions.

7.5 Toss Winner and Match Winner

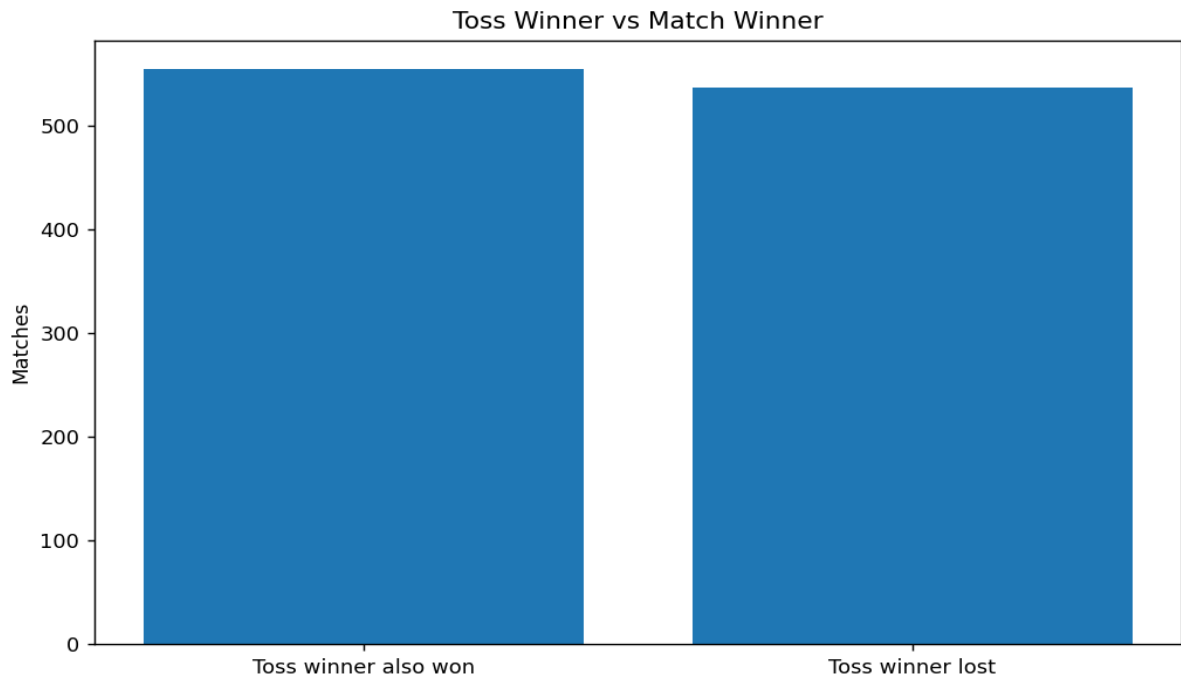


Figure 7.5: Toss Winner and Match Winner

The toss winner also won 50.83% of matches with a recorded winner. This is a descriptive association and does not by itself establish causation.

7.6 Match Result Types

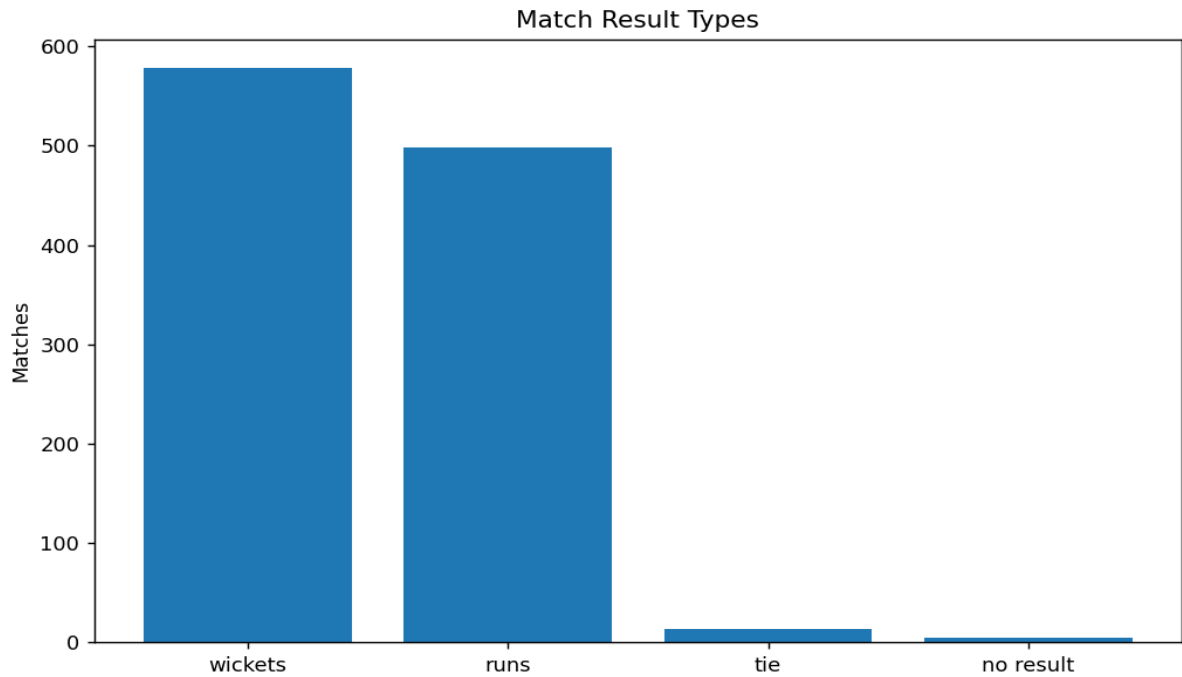


Figure 7.6: Match Result Types

Wins by wickets are the most frequent recorded result type, with 578 matches, while wins by runs account for 498. Ties and no-result matches are also represented.

7.7 Top Venues

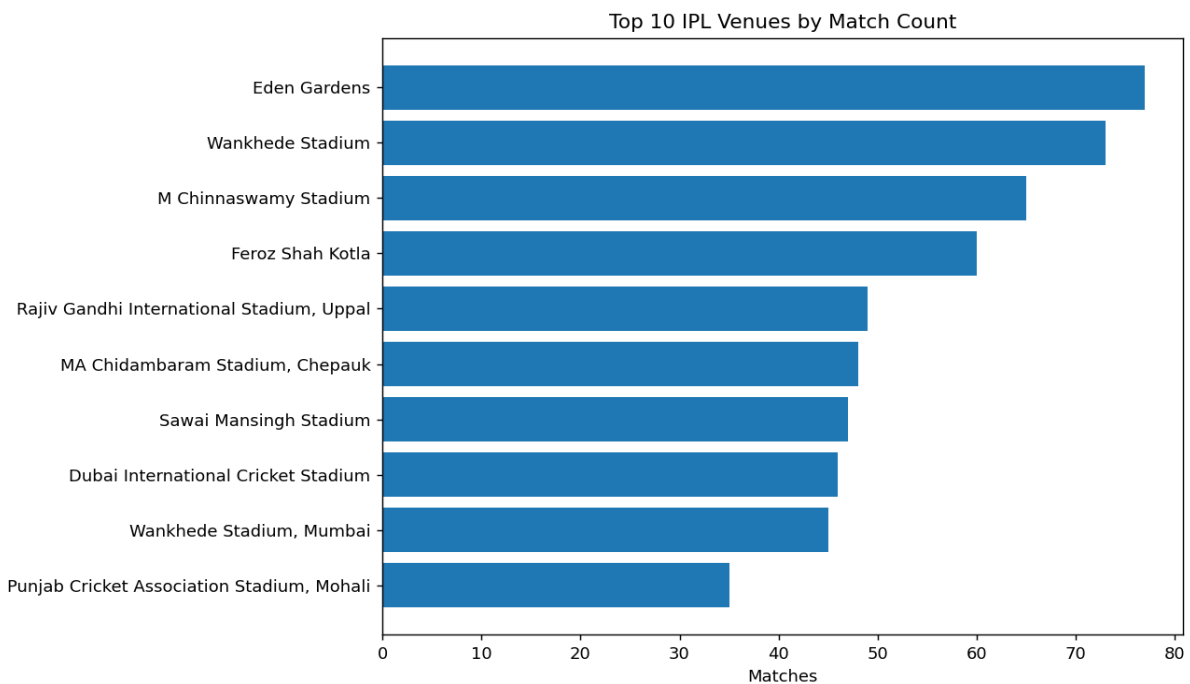


Figure 7.7: Top Venues

The venue chart shows the ten venues with the highest number of matches. Eden Gardens has the highest count in the analyzed dataset.

7.8 Winning Margin Distribution

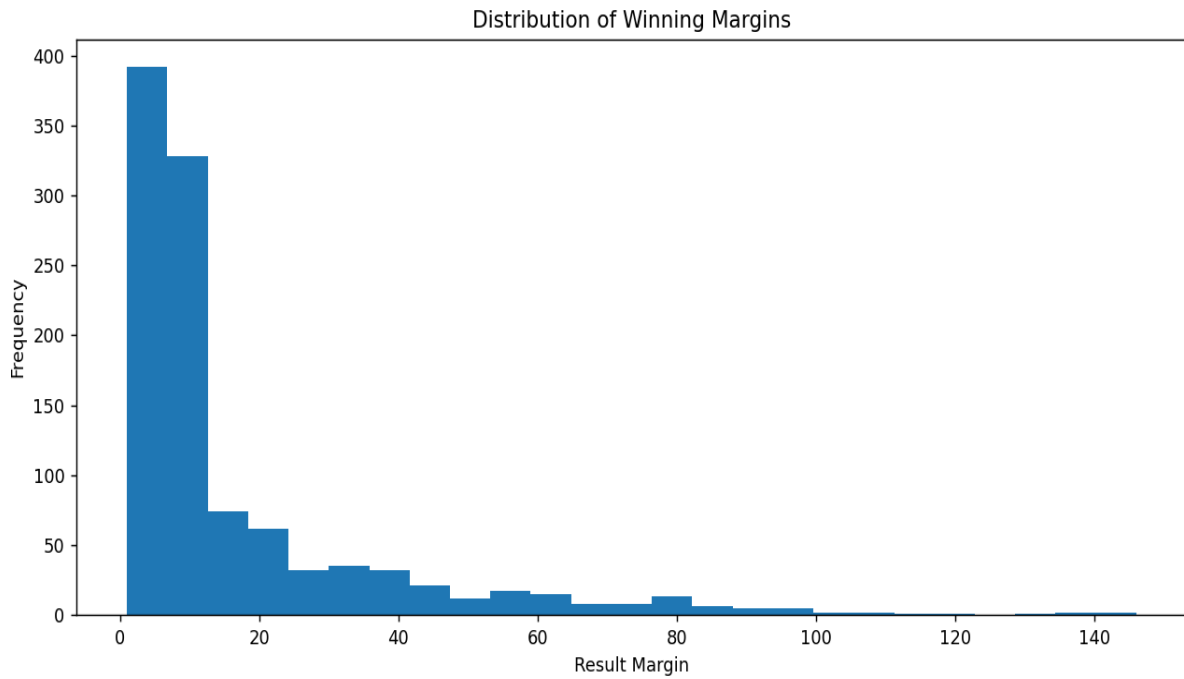


Figure 7.8: Winning Margin Distribution

The distribution shows how recorded winning margins are spread across matches. Very large margins occur less frequently than smaller or moderate margins.

7.9 Average Winning Margin by Season

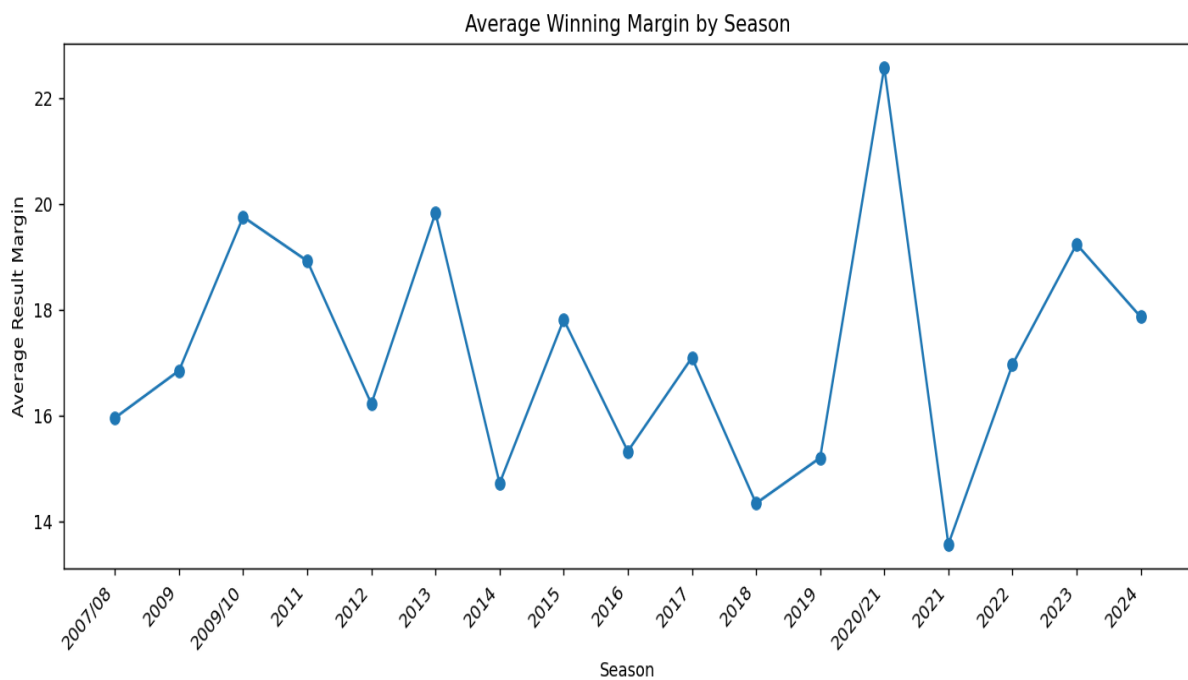


Figure 7.9: Average Winning Margin by Season

This line chart compares the average recorded result margin across season labels. Changes between seasons indicate differences in the distribution of match-winning margins.

7.10 Correlation Matrix

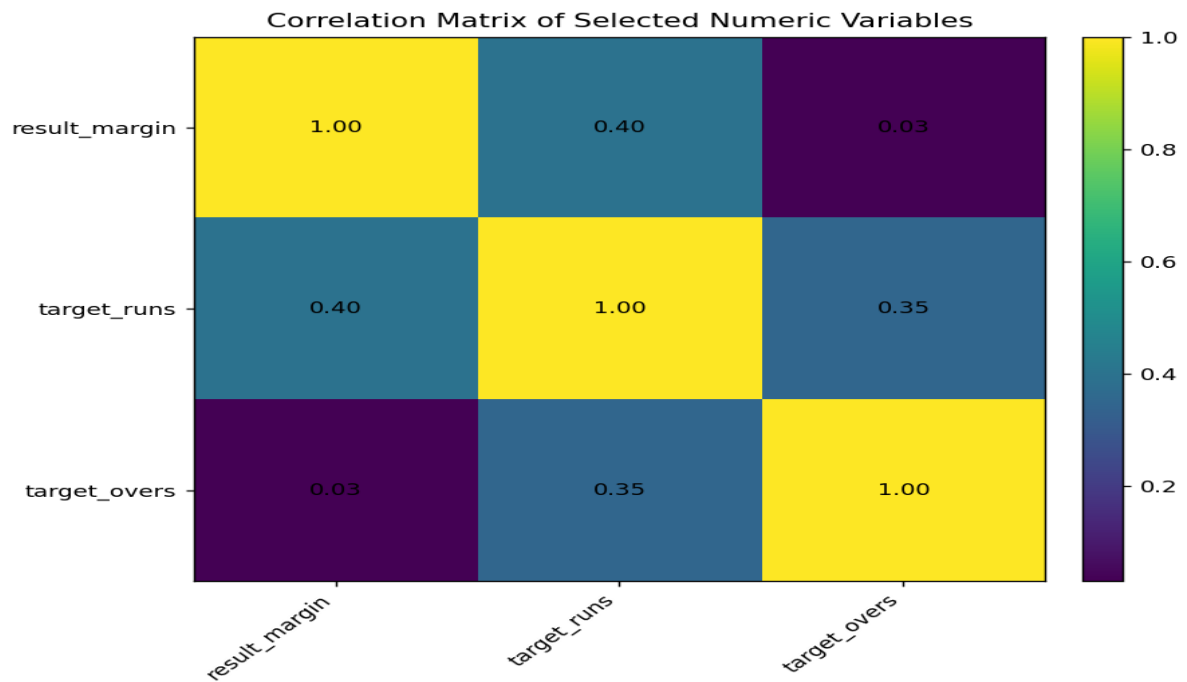


Figure 7.10: Correlation Matrix

The correlation matrix summarizes linear relationships among selected numeric variables. Correlation describes association and should not be interpreted as causation.

7.11 Runs vs Wickets

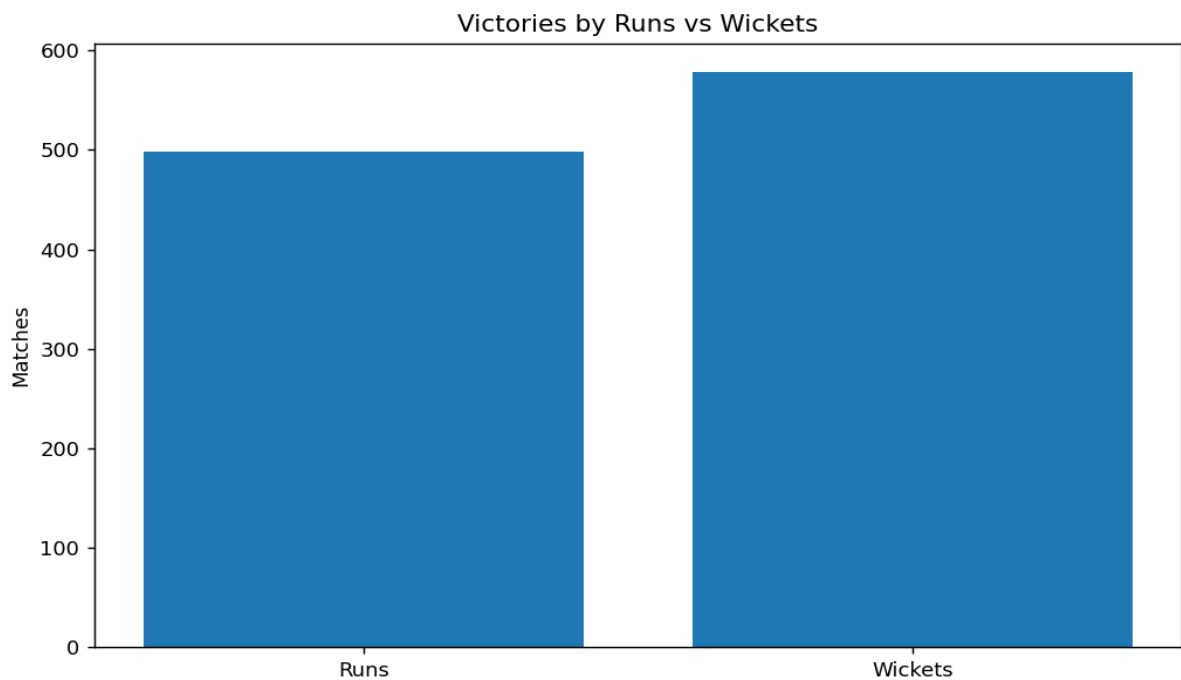


Figure 7.11: Runs vs Wickets

The dataset contains 578 wicket-margin wins and 498 run-margin wins, showing the relative frequency of successful chases and successful defenses in the recorded match results.

7.12 Special Outcomes and Conditions

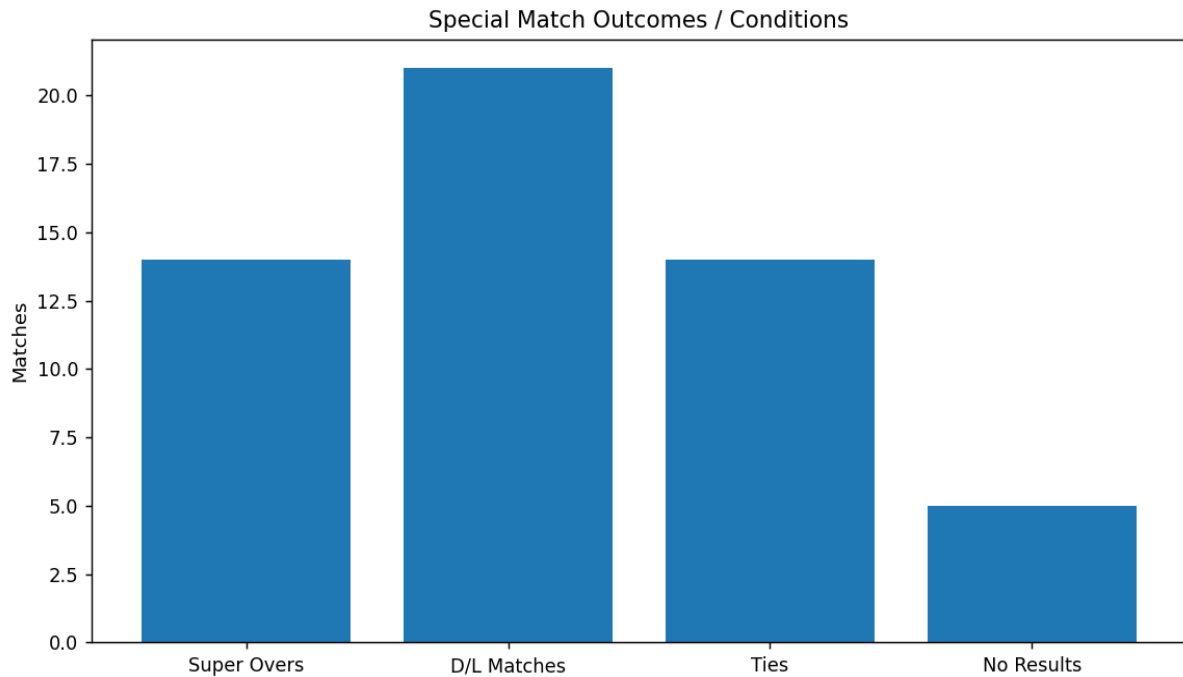


Figure 7.12: Special Outcomes and Conditions

The dataset includes 14 super-over matches, 21 D/L matches, 14 ties and 5 no-result matches. These are relatively uncommon compared with the full match count.

8. Key Findings

- The dataset contains 1,095 matches and 20 variables across 17 season labels.
- Mumbai Indians has the highest recorded win count in the analyzed data, with 144 wins.
- Mumbai Indians has the highest number of match appearances, with 261 appearances.
- Eden Gardens is the most represented venue, with 77 matches.
- The toss decision was to field in 704 matches and to bat in 391 matches.
- The toss winner also won the match in 50.83% of matches with a recorded winner.
- Matches won by wickets (578) outnumber matches won by runs (498) in this dataset.
- There were 14 ties and 5 matches without a recorded winner.
- The dataset contains 14 super-over matches and 21 matches with a recorded D/L method.
- The largest recorded run-margin victory is 146 runs, while the largest recorded wicket-margin victory is 10 wickets.

9. Limitations

- The analysis is descriptive and does not establish causal relationships.
- Historical team names require standardization for longitudinal comparisons.
- Some fields are missing for specific match circumstances.
- Source season labels are retained as provided.
- Ball-by-ball and player-level details are outside the scope of this match-level dataset.

10. Conclusion

The EDA summarizes 1,095 IPL matches and highlights patterns in participation, wins, venues, toss decisions and match outcomes. 578 matches were wicket-margin victories and 498 were run-margin victories. The toss winner also won 50.83% of matches with a recorded winner. The project demonstrates how descriptive analysis and visualization can make a large sports dataset easier to interpret.

11. Future Scope

- Add ball-by-ball and player-level analysis.
- Create an interactive dashboard.
- Analyze detailed batting, bowling and fielding performance.
- Apply predictive modeling after the EDA stage.
- Compare home, away and neutral venue performance.

12. Tools Used

Python, Jupyter Notebook, Pandas, NumPy and Matplotlib.